

NON-GONOCOCCAL URETHRITIS

Non-gonococcal urethritis (NGU) is one of the most common STIs affecting men. Typically, it features inflammation of the urethra, with or without a discharge of pus; inflammation and soreness at the end of the penis; and pain on passing urine, particularly when the urine is concentrated first thing in the morning. In about half of all cases, the agent responsible is *Chlamydia trachomatis*, a bacterium that can also infect women, leading

to chlamydial infection. Other possible causes of NGU include the bacterium *Ureaplasma urealyticum*; the protozoan *Trichomonas vaginalis*; the fungus *Candida albicans*; the genital warts virus (human papillomavirus, HPV); and the genital herpes viruses. It is important for both partners to seek treatment to avoid reinfecting each another. To prevent STIs, sexually active people should limit their sexual partners, and use a condom for penetrative sex.

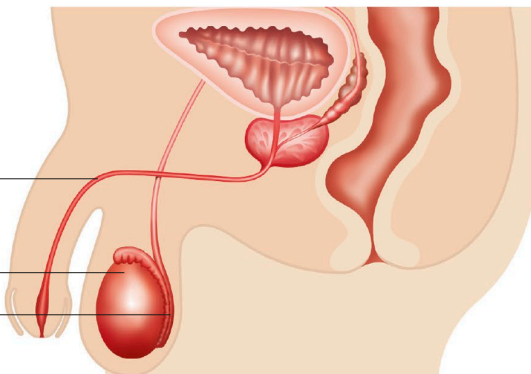
SYMPTOMS OF NGU

Inflammation of the urethra causes pain and soreness at the external opening on the penis, and painful urination. If the infection spreads, the epididymides and the testes may also become swollen.

Urethra
Inflammation of urethra causes pain on passing urine

Testis
May become swollen if infection spreads

Epididymis
Sometimes also becomes inflamed



SYPHILIS

Syphilis can affect both men and women. It is caused by *Treponema pallidum*, a bacterium that enters the body via the genitals. It first affects the organs of reproduction, and spreads to other parts of the body. An infectious sore (chancre) appears on the penis or vagina, lymph nodes swell, and then a rash and wart-like patches develop on the skin. With no treatment, it can proceed to a final, possibly fatal, stage characterized by personality changes, mental illness, and nervous system disorders. Today, the disease rarely progresses to this stage.

CHLAMYDIAL INFECTION

Chlamydial infection is a very common STI and occurs only in women. It is caused by *Chlamydia trachomatis*, which inflames the reproductive organs, and causes symptoms including vaginal discharge, a frequent urge to urinate, lower abdominal pain, and pain during intercourse. Chlamydial infection can lead to PID (see opposite), if left untreated, and may then cause infertility.

BACTERIA IN CERVICAL SMEAR

This micrograph (x400) of a cervical smear shows *Chlamydia trachomatis* bacteria (pink cells within large blue cell).

